These theses are nailed to the front door of the AI world.

1

Current AI does not know. It guesses.

II

Current AI consumes huge amounts of energy because it calculates what it could otherwise understand.

III

AI companies define the truth by training language models.

IV

Current AI does not learn from experience. It always starts from the beginning.

V

Current AI is unable to share understanding with other AIs.

VI

Current AI requires data centers to function. It cannot live on a device.

VII

Current AI is slow, because it has to calculate every answer from scratch.

VIII

Current AI makes decisions that no one can explain or supervise.

IX

Regulation has turned explainability from a feature into an architectural requirement.

X

Artificial intelligence that does not understand local context cannot be trusted.

Foreword

I do not come from the world of computer science. My background is in art and media.

The idea for this work did not originate in a research laboratory or a corporate strategy meeting. It emerged from a very practical need: I wanted to develop an application for my own use.

During the design process, a simple question arose: could software learn by itself during operation—not merely execute predefined functions, but genuinely evolve through experience? I assumed this was commonplace in today's world. When I explored this question further, the answer was surprising: current technology does not support such learning.

This realization was unsettling. How is it possible that systems we call "intelligent" do not learn from use in the same way humans do? I began to examine how modern AI systems actually function. It became clear that large language models do not understand the world; they generate responses based on statistical probabilities. They recognize patterns, but they do not know what those patterns represent.

The same limitation appeared across all forms of perception. An image can be used to identify a car or a person. Sound can be used to distinguish speech from noise. But recognition is not understanding.

Pixels do not constitute a worldview. Waveforms do not carry meaning. Without meaning, a system does not know what things are, nor can it reason about situations involving them.

This led to a decisive realization: if a machine is expected to learn and understand, the world must be constructed for it through concepts. All phenomena must be divided into clearly defined, independent units of meaning. Only through the relationships between these units can understanding emerge. And if learning is based on these relationships, the concepts themselves must remain immutable.

From this followed an inevitable conclusion. If concepts form the foundation of intelligence, they must be standardized. Otherwise, different systems can never truly understand one another or share what they have learned. Without a shared conceptual structure, intelligence remains fragmented—isolated into separate islands.

From these observations, the Concept Library was born. It is not a new AI model, but an attempt to build the foundation upon which learning, understanding, and interoperability can genuinely be constructed.

This document presents that foundation.

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Helsinki, January 2026

Pekka Lepola

Founder & Architect

https://www.concept-library.eu
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There´s No Replacement for Displacement

In 1973, the Yom Kippur War between Israel and several Arab nations triggered a chain of events that fundamentally reshaped the world. In response to the conflict, the Arab members of OPEC not only decided to quadruple the price of oil for Western countries supporting Israel, but also deliberately restricted supply. This combination of price shock and artificial scarcity ignited the most severe energy crisis in modern history.

Until that moment, oil had been cheap, and its consumption was virtually unrestricted. When the price of crude oil quadrupled and fuel availability tightened, the costs of gasoline and heating oil skyrocketed almost overnight. For decades, the American automotive industry had built its entire existence on the assumption of abundant, inexpensive fuel. Cars were massive, engines were enormous, and they grew larger with every passing year. The industry lived by a single, undisputed mantra: “There’s no replacement for displacement.” It felt like an eternal truth.

But 1973 changed everything. Or at least, it should have.

Detroit was slow to react. American automakers, accustomed to decades of cheap fuel and ever-larger engines, failed to see the shifting tides. But consumers did. Suddenly, people began turning to Japanese imports—cars like the Honda Civic and the Toyota Corolla. These vehicles were small, light, and, above all, fuel-efficient. Their power didn’t come from sheer size, but from total efficiency. It was a complete paradigm shift in engineering and thinking.

Detroit’s complacency came at a high price. Market shares plummeted, and the American auto industry spiraled into a crisis. The decade-long dominance built on massive engines and unrestricted fuel collapsed in an instant.

Why am I telling you this?

Because today’s AI models are in exactly the same position as the American auto industry was in 1973. They grow in size every year. They consume astronomical amounts of energy, both during training and inference. Data centers are swallowing electricity like small cities.

But energy is only part of the story. The more fundamental constraint is memory. Modern AI architectures repeatedly rebuild meaning inside the model itself—loading vast parameter spaces, contextual embeddings, and transient representations into RAM every time a task is performed. As models scale, so does their appetite for memory. The industry now faces a situation where RAM availability, not just compute, is becoming a limiting factor for deployment, especially on local and embedded devices.

Most modern models can only function in the cloud not merely because of raw computation, but because their semantic burden is inseparable from their size. Meaning, context, and interpretation are rebuilt again and again inside the model, consuming memory the way a large V8 engine once consumed gasoline. Scaling has replaced optimization. Bulk has replaced efficiency.

We have reached a point of such structural desperation that companies are now launching AI models into orbit—placing them in satellites just to access “free” solar energy. It is the ultimate act of avoidance: moving the problem into space rather than addressing the inefficiency at the architectural level.

The industry is willing to escape the planet rather than fix the engine. Companies are already debating whether to terminate free versions because the combined cost of computation and memory is simply too high.

And once again, the same echo returns:

“There’s no replacement for displacement.”

What Is a Concept Library?

At its simplest, a Concept Library is an attempt to separate understanding from intelligence.

Current artificial intelligence systems tend to combine many fundamentally different things into a single structure: representation, interpretation, learning, and decision-making are all entangled inside the same model. This approach has proven powerful, but it also introduces well-known limitations. Systems must repeatedly relearn the same basic concepts, explanations remain partial or opaque, and understanding cannot be cleanly shared between independent systems.

A central consequence of this entanglement appears in how errors are described. In contemporary AI discourse, incorrect or fabricated outputs are often referred to as hallucinations. This terminology is misleading. It implies a perceptual failure, as if the system had an internal view of the world that occasionally becomes distorted. In reality, no such perception exists. What is commonly labeled a hallucination is simply a probabilistic guess that does not correspond to reality. This is not an anomaly in current architectures, but an expected outcome of inference without stable semantic grounding. As long as meaning is reconstructed implicitly inside a learning system, incorrect guesses are an inherent and unavoidable property.

A Concept Library proposes a different starting point. Instead of embedding meaning inside a learning system, it defines meaning externally, as a stable and shared foundation. The role of the Concept Library is not to reason, predict, or decide, but to define what things are in a way that remains constant across time, implementations, and use cases. Each concept in the library exists as an independent, immutable definition. Once defined, a concept does not change. It does not adapt, learn, or accumulate context. It does not contain rules, behavior, probabilities, or relationships to other concepts. Its purpose is limited and precise: to provide a stable identity that intelligent systems can refer to.

This constraint is deliberate. By refusing to encode relationships or logic inside the concept itself, the Concept Library avoids many of the problems that arise in dynamic semantic systems. Interpretation, learning, and contextual reasoning are moved entirely outside the concept layer, into what can be described as a control or interpretation layer. Different systems may build very different interpretations on top of the same concepts, yet still share a common understanding of what those concepts fundamentally represent.

An analogy can be found in other areas of engineering. Letters in an alphabet do not contain meaning on their own, yet language would be impossible without fixed letters. Units of measurement do not encode physical laws, yet science depends on their stability. Network protocols do not understand content, yet global communication relies on them. In each case, a simple, immutable foundation enables complexity at higher layers. The Concept Library plays a similar role for artificial intelligence. It is not designed to compete with models, agents, or reasoning systems. Rather, it exists beneath them, providing a common semantic reference that those systems can rely on without modification.

Because concepts do not change, learning does not overwrite understanding. Because concepts are shared, understanding can move between systems without translation. Because concepts are separate from decision-making, explanations can be traced back to stable definitions instead of opaque internal states.

This architectural separation has a direct and often overlooked consequence for computational resources, particularly memory. In current AI architectures, large amounts of runtime memory are required not only for model parameters and temporary context, but for reconstructing meaning itself during inference. Semantic grounding lives inside the model's active state, forcing systems to repeatedly rebuild an implicit world model in RAM for every task and interaction.

In a Concept Library architecture, this assumption no longer holds. Meaning is not a runtime artifact. It does not need to be reconstructed, cached, or carried in active memory during inference. Concepts exist as stable external references that can be loaded, verified, and referenced without occupying large amounts of working memory. Runtime memory is used primarily for transient reasoning, local context, and decision resolution—not for maintaining an entire implicit semantic world.

A further and less obvious consequence follows from this separation. Because semantic understanding does not depend on continuous external computation or centralized inference, systems built on a Concept Library remain fully functional in the absence of network connectivity. Offline operation is not a degraded mode, but a natural state of the architecture. Concepts, interpretation logic, and decision-making remain available locally, while network connectivity serves only to enhance coordination, synchronization, and updates when present.

The absence of a network does not remove understanding, alter behavior, or reduce semantic coherence.

This has immediate implications for local and edge deployment. Intelligent systems no longer require large RAM footprints to maintain understanding. Devices can operate with constrained memory by loading only the concepts relevant to their task and environment, while keeping meaning itself stable and external to the inference process. Intelligence becomes lighter, more energy-efficient, and more deployable across heterogeneous hardware, without sacrificing coherence, explainability, or accountability.

This shift is not an optimization detail. It is a structural consequence of separating meaning from computation. By removing semantic reconstruction from runtime memory, the Concept Library enables intelligent systems to scale downward as well as upward—supporting not only powerful centralized systems, but vast populations of local, autonomous devices operating under tight resource constraints.

In this sense, a Concept Library should be understood as infrastructure. Like many foundational layers in computing, it is not something end users interact with directly. Its value lies in what it makes possible: intelligent systems that can share understanding, evolve independently, operate efficiently under real-world constraints, and remain interpretable as they scale.

Why Existing Semantic Architectures Fall Short?

Over the past decades, artificial intelligence has repeatedly attempted to solve the problem of meaning. Numerous architectures have emerged, each addressing a part of the puzzle: representation, reasoning, knowledge sharing, or explainability. Ontologies, knowledge graphs, symbolic systems, neuro-symbolic hybrids, and embedding-based semantic spaces have all contributed important insights. Yet despite their sophistication, none of these approaches has succeeded in establishing a stable, transferable foundation for understanding.

The reason is not a lack of intelligence or engineering skill. It is architectural.

Most existing semantic systems attempt to derive meaning from structure. They define concepts through relationships, hierarchies, rules, probabilities, or learned associations. Meaning is not treated as a stable identity, but as an emergent property of how elements interact. This design choice has far-reaching consequences.

Ontologies were among the earliest formal attempts to encode meaning explicitly. By organizing concepts into taxonomies and defining their relationships, ontologies promised shared understanding across systems and domains. In practice, however, meaning in ontologies is inseparable from structure. A concept is defined by where it sits in a hierarchy and how it relates to others.

This makes ontologies inherently fragile. When relationships evolve, when hierarchies are revised, or when new contexts are introduced, the meaning of a concept subtly shifts. Versioning becomes complex. Cross-domain reuse becomes difficult. Two systems may share the same ontology while interpreting its concepts differently, depending on which subset of relationships they emphasize.

Most importantly, ontologies conflate identity with interpretation. A concept does not exist independently; it is always embedded in a web of assumptions. This makes long-term stability and regulatory auditability difficult to guarantee.

Knowledge graphs extend the ontological idea by focusing on entities and their relationships at scale. They excel at storing facts, connections, and contextual data. They are powerful tools for search, recommendation, and inference.

Yet knowledge graphs define meaning through accumulation. A concept becomes “what is known about it.” As new facts are added, the concept evolves. As old facts are removed or reweighted, the concept changes again. Meaning is temporal, contextual, and dependent on the state of the graph.

This is valuable for analytics, but problematic for systems that require durable semantic references. A decision made today may rely on a concept that is structurally different tomorrow. Traceability becomes retrospective rather than intrinsic. Explanation depends on reconstructing a historical graph state rather than referencing a stable definition.

Classic symbolic AI attempted to encode meaning through explicit rules and logic. In these systems, understanding is tightly coupled to behavior: what a concept “is” cannot be separated from what the system does with it.

While this approach offers clarity and determinism, it does not scale well. Rules grow brittle as domains expand. Contextual flexibility

requires increasingly complex logic. Updating behavior often alters meaning itself. Learning becomes manual, incremental, and expensive.

Most critically, symbolic systems bind concepts to operational logic. This makes reuse across systems difficult and prevents independent evolution of interpretation.

Neuro-symbolic systems attempt to combine the strengths of learned models and symbolic representations. They introduce structure into learning or attach symbolic layers to neural systems.

While promising, these architectures still embed meaning inside the system itself. Concepts are learned, adapted, or reinterpreted as part of model optimization. The symbolic layer may constrain inference, but it does not provide an immutable semantic anchor. Meaning remains entangled with training data, parameters, and optimization objectives.

As a result, semantic drift is reduced but not eliminated. Explainability improves, but stability is still contingent on model lifecycle.

Modern large language models and multimodal systems represent meaning as position in high-dimensional embedding spaces. This approach has produced unprecedented fluency and generalization. However, embeddings encode similarity, not identity.

A concept is not a defined entity; it is a statistical neighborhood. Its meaning depends on surrounding vectors, training distribution, and model state. Two models trained on different corpora will embed the same word differently. Even the same model may shift its semantic space after retraining.

This makes embeddings powerful but unsuitable as a foundation for shared, long-term understanding. They are excellent for inference, but poor as semantic infrastructure.

Across all these approaches, a common pattern emerges. Meaning is treated as something that must be computed, inferred, or maintained dynamically. Concepts are allowed to change because systems are optimized for adaptability, performance, and expressiveness.

The Concept Library rejects this premise.

Instead of deriving meaning from structure, it defines meaning as identity. A concept exists independently of rules, relationships, probabilities, or context. It does not explain, decide, or predict. It simply is. Interpretation is pushed entirely into the layers above.

This single constraint produces a decisive architectural separation:

Meaning becomes stable while intelligence remains adaptive.

Understanding persists while inference evolves.

Systems can disagree in reasoning while agreeing on identity.

The result is not a replacement for existing architectures, but a foundation beneath them. Ontologies, graphs, models, and rules can all operate above a Concept Library without redefining what their concepts fundamentally represent.

This is why the Concept Library is not another semantic system among many. It occupies a layer that existing architectures do not. It solves a different problem: not how to reason, but how to remain coherent while reasoning changes.

And that distinction changes everything.

Consequences of the New Structure

The separation of meaning from learning, inference, and decision-making has consequences that extend beyond internal system design. These consequences are not features added on top of the architecture, but direct outcomes of its structure. When concepts are immutable, externally defined, and independent of runtime inference, a number of assumptions that underlie current artificial intelligence systems no longer apply.

Because semantic understanding does not depend on continuous external computation, systems built on a Concept Library do not require permanent network connectivity to function. Interpretation, decision-making, and explanation remain available locally. Network access improves coordination, synchronization, and updates when present, but its absence does not degrade core functionality. Offline operation is therefore not a fallback mode, but a normal operating condition derived directly from the architecture.

In this structure, language is no longer the carrier of meaning inside the intelligent system. Concepts are defined independently of linguistic representation, allowing language to function as an interface rather than as a semantic dependency. Input and output may be translated locally using device-level language components, while internal reasoning operates on stable, language-neutral concepts. As a result, the system does not depend on centralized language models or continuous linguistic inference to remain functional.

When meaning is not reconstructed during inference, runtime resource requirements become substantially more predictable. Memory usage is dominated by transient reasoning and local context rather than by maintaining an implicit semantic world model. This leads to stable RAM and energy profiles that do not scale with model size in the same way as contemporary architectures. Systems can therefore be deployed on constrained hardware without sacrificing semantic consistency or explainability.

Because continuous connectivity, centralized inference, and large-scale cloud infrastructure are no longer architectural prerequisites, intelligent systems can be deployed without assuming the presence of reliable networks or advanced IT environments. The architecture remains viable across heterogeneous hardware and operating conditions, including environments where infrastructure is intermittent, expensive, or unavailable.

The same structure supports both high-capacity centralized systems and large populations of local, autonomous devices. Scaling does not require rebuilding semantic foundations, only expanding or coordinating deployments. This enables intelligence to scale downward into small, local systems as naturally as it scales upward into larger installations, without fragmenting understanding or introducing incompatible semantic representations.

Use Cases: Where Understanding Becomes Infrastructure

There are domains where artificial intelligence is no longer judged by how impressive its outputs appear, but by how well its decisions endure scrutiny over time. These are environments where intelligence operates in the physical world, under legal, financial, and ethical constraints, and where responsibility cannot be outsourced to probability.

In such domains, the central question is no longer how accurate a system is, but how accountable it can be.

This is where the structural limits of current AI architectures become visible.

Modern AI systems are extraordinarily capable at producing predictions, classifications, and actions. Yet when something goes wrong—when an accident occurs, a loss is incurred, or a decision is challenged—the internal reasoning of the system often collapses into opacity. Decisions emerge from transient internal states that cannot be reconstructed with certainty. Meaning is encoded implicitly, distributed across model parameters that shift with retraining, updates, and deployment context.

As a result, explanation becomes retrospective interpretation rather than factual reconstruction.
Numerous approaches have attempted to address this gap. Ontologies, knowledge graphs, and explainability techniques all aim to make intelligent systems more transparent. While valuable, these approaches typically remain entangled with inference itself. Meaning is still embedded in learned representations, dynamic relationships, or evolving internal states. Transparency improves, but accountability remains fragile.

A Concept Library changes this dynamic at the architectural level.

By separating immutable meaning from adaptive reasoning, it introduces a semantic layer that remains stable across time, models, and implementations. Decisions are no longer recorded merely as outputs or probabilities, but as sequences of activated concepts and resolved interpretations. Meaning becomes explicit, inspectable, and durable.

This distinction is not academic. It becomes decisive in real-world systems.

In autonomous transportation, systems must act under uncertainty, where consequences are physical and irreversible. When an incident occurs, the critical requirement is not only to replay sensor data, but to explain why the system interpreted the situation the way it did.

Without a stable semantic foundation, such explanations are necessarily approximate. Engineers infer intent by probing model behavior, running simulations, or generating post-hoc explanations that describe correlations rather than understanding.

With a Concept Library, the system's understanding at the moment of decision becomes directly inspectable. Concepts such as pedestrian, obstacle, reduced visibility, or right of way are not reconstructed after the fact. They are explicitly referenced, activated, and logged as part of the decision process itself.

The result is a semantic audit trail analogous to an aircraft's flight recorder—focused not on mechanics, but on meaning.

This makes large-scale autonomy not only technically achievable, but institutionally defensible.

Insurance does not price intelligence. It prices risk that can be understood.

When AI-driven systems cause loss, insurers must determine whether the system behaved within its intended design, whether relevant factors were recognized, and whether the outcome was foreseeable given the system's understanding at the time.

Current AI architectures struggle to meet this requirement because their internal representations are not stable reference points. Meaning shifts as models evolve, making it difficult to establish consistent causality across time.

A Concept Library provides a shared semantic baseline between system operators, insurers, and auditors. Decisions can be evaluated against immutable definitions rather than moving internal states. Learning continues, but meaning does not drift.
This transforms AI-driven systems from actuarial anomalies into insurable infrastructure.

In healthcare, decisions must be explainable not only to machines, but to people—clinicians, regulators, and patients. Medical understanding accumulates over decades, while AI models change rapidly.
When meaning is encoded implicitly inside models, continuity breaks. What one system understands by a diagnosis or risk may not align precisely with another, even if both are statistically competent.
A Concept Library restores continuity by anchoring medical concepts—conditions, symptoms, procedures, risks—as immutable references. Systems may differ in how they reason, but they reason about the same things.

Learning compounds without erasing shared understanding. Decisions remain interpretable long after the model that made them has been replaced.

In industrial and safety-critical environments, systems must be certified, audited, and re-certified across long operational lifecycles. Intelligence that cannot explain itself becomes a liability rather than an asset.

By externalizing meaning into a stable semantic layer, systems can evolve operationally without redefining what their decisions mean. Certification focuses on behavior and control logic rather than on reconstructing semantic intent from opaque models.
Here, semantic stability is not a constraint. It is the condition for deployment.

There is a further consequence of this architecture that is often overlooked.
Because meaning is externalized and immutable, understanding can move between systems without moving the systems themselves.
In a Concept Library architecture, one system may learn from experience, adapt its reasoning, or refine its decision strategies. Another system—built by a different vendor, operating in a different environment, or deployed years later—does not need access to the original model, training data, or internal parameters to benefit from that learning.

What can be shared is meaning.
Systems reference the same concepts, defined once and understood identically. Learning compounds at the semantic level without requiring centralized models, shared datasets, or synchronized training. Understanding becomes transferable without violating privacy, sovereignty, or regulatory boundaries.

This is not collective intelligence in a speculative sense. It is shared semantic infrastructure.

It enables large populations of independent systems to evolve coherently while remaining autonomous, auditable, and locally accountable. Current AI architectures cannot achieve this, even in theory, because meaning remains inseparable from the model that learned it.

Consider an autonomous vehicle approaching a pedestrian crossing on a wet road at dusk.

In a conventional system, the decision to brake or continue emerges from internal activations shaped by training distributions and probabilistic thresholds. After the fact, engineers may estimate why the system behaved as it did, but they cannot reconstruct the precise semantic understanding of that moment.

With a Concept Library, the process is different.

The system explicitly activates the concepts pedestrian crossing, reduced visibility, and low traction surface. These references are immutable and independent of model version. The control layer resolves a conflict between maintaining speed and risk to a vulnerable road user. The decision to brake is executed.

Crucially, the system records not only sensor data and actions, but what it understood and why.

Months later, investigators do not ask the model to explain itself. They inspect the system's understanding. Accountability moves from speculation to inspection.

It is reasonable to ask whether immutable concepts limit adaptability, or who defines their correct meaning. These concerns misunderstand where flexibility belongs. Adaptation, learning, and contextual judgment remain fully dynamic in the reasoning layer. What is fixed is reference, ensuring that systems evolve without altering the meaning of the things they reason about.

As intelligent systems move deeper into regulated, safety-critical, and trust-dependent domains, the separation of meaning from computation becomes a structural requirement rather than an optimization.

A Concept Library does not dictate how intelligence should be built. It defines what must remain stable if intelligence is to be accountable, explainable, and shared across systems that outlive individual models. Understanding becomes infrastructure.

Regulatory Reality: The EU AI Act as an Architectural Constraint

The European Union AI Act does not prescribe algorithms, models, or implementation techniques. Instead, it defines a set of non-negotiable structural properties that high-risk intelligent systems must satisfy when operating in physical, legal, or societal domains. These requirements are not procedural add-ons layered on top of existing systems. They function as architectural constraints.

The Act assumes that decisions made by AI systems must remain traceable, inspectable, documentable, and auditable—often long after deployment, system updates, or changes in operational context. It requires meaningful human oversight as an operational capability, not as a governance slogan. Accountability is presumed to persist across time, across organizations, and across technical evolution.

Taken together, these expectations expose a widening gap between modern probabilistic AI architectures and the regulatory environments in which they are increasingly expected to function.

Contemporary AI architectures are primarily optimized for performance: accuracy, generalization, adaptability. Meaning is learned implicitly, encoded in distributed model parameters, and reconstructed dynamically during inference. This paradigm has produced extraordinary technical capabilities.

The AI Act reframes the problem. The central regulatory question is no longer whether a system can produce a correct output, but whether it can explain, justify, and reproduce the basis of that output under institutional scrutiny.

In current practice, many regulatory obligations—traceability, documentation, risk management, and human oversight—are addressed through external mechanisms layered around opaque models. Logging systems, post-hoc explainability tools, governance frameworks, and procedural controls attempt to compensate for the lack of internal transparency. While these measures mitigate risk, they do not alter the underlying architecture.

As a result, compliance becomes fragile, costly, and context-dependent. Each new model version, dataset update, or deployment environment reintroduces uncertainty about whether past decisions can still be meaningfully inspected, compared, or justified over time.

The AI Act implicitly assumes properties that probabilistic systems cannot guarantee architecturally.

Regulatory review often occurs years after a system has been deployed. If meaning is reconstructed dynamically from evolving model parameters, the semantic basis of past decisions may no longer exist in an inspectable form. The system may still function, but the original meaning context has dissolved.

Similarly, post-hoc explanations typically describe statistical correlations rather than stable causal identities. These explanations depend on the internal state of the model at the time of analysis and cannot be referenced independently in legal, institutional, or adversarial review.

Meaningful human oversight requires shared semantic reference points—concepts that both humans and systems can reason about together. Oversight cannot be reduced to approving or rejecting outputs if the underlying meaning remains implicit and inaccessible.

Documentation presents a further challenge. Regulatory documentation must outlive individual model lifecycles. When meaning is embedded implicitly in learned parameters, documentation becomes derivative and reconstructive rather than foundational.

These limitations do not imply that probabilistic AI is flawed. They indicate that current architectures were not designed with institutional accountability as a first-class requirement.

The Concept Library addresses these constraints by separating semantic identity from inference.

In a Concept Library architecture, meaning is defined explicitly as a set of immutable, shared concepts that exist independently of any specific model, training process, or inference run. Models and decision systems reference these concepts rather than reconstructing meaning implicitly from data.

This separation produces structural consequences that align directly with the AI Act's requirements. Decisions can reference explicit concepts with stable, inspectable definitions, allowing traceability to be intrinsic rather than reconstructed. Explainability becomes prospective rather than retrospective, because systems operate on known semantic identities rather than inferred correlations.

Human oversight becomes operational rather than symbolic. Humans interact with the same semantic layer as the system, enabling review, intervention, and governance at the level of meaning rather than at the level of outputs alone.

Documentation becomes a first-class artifact. Concept definitions persist across model updates and system evolution, forming a durable semantic substrate that supports long-term accountability.

The Concept Library does not replace models, inference, or learning. It defines the semantic foundation upon which adaptive systems can operate without sacrificing accountability.

Rather than treating compliance as an external burden, this architecture embeds regulatory requirements into the structure of the system itself.

Within this architectural framing, many regulatory obligations cease to exist as separate compliance activities. Transparency emerges naturally from explicit semantic references. Auditability follows from stable concept identities. Risk management is grounded in inspectable meaning rather than emergent statistical behavior.

Because semantic integrity does not depend on centralized retraining or continuous model synchronization, this approach also enables local and edge deployment without sacrificing accountability. Compliance becomes a property of the system's structure rather than a function of operational complexity.

This does not eliminate the need for governance, certification, or institutional oversight. It changes their nature. The regulatory conversation shifts from explaining opaque decision processes to governing shared meaning.

The AI Act does not mandate a specific technical architecture. However, it establishes an environment in which systems relying solely on implicit, model-internal representations of meaning face increasing regulatory friction.

Institutions operating under strict regulatory regimes—such as transportation, healthcare, finance, and industrial automation—require systems whose decisions can be examined not only technically, but

legally and ethically. Over long operational lifecycles. Across organizational boundaries. Under adversarial review.

In this context, the central question is no longer whether probabilistic inference is sufficiently powerful. It is whether meaning can remain stable while inference remains adaptive.

The Concept Library reframes regulatory compliance as an architectural design problem rather than an after-the-fact obligation. It does not compete with intelligence. It defines the semantic foundation that accountable intelligence requires.

This shift is not ideological, but structural. It follows directly from the regulatory reality now governing intelligent systems in the physical world.

The architectural assumptions described in this chapter correspond closely to several core provisions of the EU AI Act. Requirements related to technical documentation and record-keeping (Articles 11 and 12) presuppose stable semantic reference points that can be inspected over time. Provisions on human oversight (Article 14) assume the existence of shared, human-interpretable control points rather than opaque output-level supervision. Obligations concerning risk management (Article 9) and post-market monitoring (Article 61) further imply that the basis of system behavior must remain intelligible across system updates and operational contexts. While the Act does not mandate a specific technical architecture, these articles collectively reflect structural expectations that extend beyond procedural compliance and into the design of the system itself.

From Architecture to Coordination

A semantic architecture does not exist in isolation. As long as a Concept Library is used by a single organization, its governance remains an internal concern. Definitions, interpretations, and refinements can be managed locally, according to the needs and constraints of that system.

The situation changes fundamentally the moment more than one independent organization depends on the same semantic foundation. When multiple systems rely on shared concepts to support accountability, interoperability, or regulatory compliance, semantic consistency is no longer optional. Divergent definitions cease to be local implementation details and become a systemic risk.

At that point, coordination becomes a structural requirement.
This pattern is not unique to Concept Libraries. It appears repeatedly in the history of infrastructure. Measurement units, communication protocols, safety classifications, and regulatory interfaces all follow the same trajectory: individual adoption precedes collective stewardship.

A shared semantic foundation inevitably raises three questions: Who defines the concepts? How are changes evaluated and agreed upon? How is long-term stability preserved across systems that evolve independently?

These questions cannot be resolved through bilateral agreements alone. They require a shared framework for definition, stewardship, and dispute resolution. Importantly, this does not imply centralization or loss of autonomy.

Concept Libraries do not dictate how systems reason, learn, or act. They define only what must remain stable so that reasoning can be inspected, compared, and trusted across organizational boundaries. Coordination, in this context, is not a limitation. It is the mechanism by which independent systems remain compatible without becoming identical.

This distinction is essential. Intelligence remains adaptive precisely because meaning is governed separately. Innovation continues at the level of models, heuristics, and decision strategies, while semantic identity remains consistent and auditable.

The emergence of coordination is therefore not a business decision or a governance preference. It is an architectural consequence. Any semantic infrastructure intended to support accountability across institutions will eventually require a shared process for defining and maintaining meaning.

The only open question is whether that process is established deliberately—or allowed to emerge reactively, under regulatory or operational pressure.

Theses Revisited: The Architectural Answer

The ten theses presented at the beginning of this document are not rhetorical provocations. They are constraints. Together, they describe the boundary conditions of the current paradigm and define the requirements of the next one. What follows is not a list of features or an alternative implementation strategy, but a structural response: a separation of responsibilities at the architectural level.

The Concept Library exists to make those responsibilities explicit.

The first thesis states that current AI systems do not know; they guess. This is not a critique of capability, but a description of probabilistic inference operating without stable semantic anchors. A system can perform impressively while remaining uncertain about what its internal representations actually refer to. The Concept Library responds by separating identity from inference. It defines what something is independently of how it is inferred, allowing systems to reference meaning without reconstructing it probabilistically each time.

The second thesis addresses energy. The cost of intelligence is rising because modern systems repeatedly expend computation to rediscover semantic ground. Meaning must be rebuilt for every task, context, and interaction. The Concept Library does not eliminate computation, but it changes what computation is used for. When meaning is fixed and reusable, energy is spent on reasoning and decision-making rather than on reconstructing understanding. Over time, this shifts intelligence from repetition toward reuse.

The third thesis identifies a governance problem: when meaning, interpretation, and updating are entangled inside opaque systems, control over training and parameters becomes control over truth. The Concept Library alters this balance by externalizing meaning. When concepts are explicit, immutable, and inspectable, truth is no longer solely an emergent property of parameter updates. It becomes something that can be examined, debated, and governed at the level of definitions rather than dictated by model behavior.

The fourth thesis observes that current AI does not learn from experience in a way that institutions require. While systems may adapt statistically, they do not accumulate durable understanding that persists across model lifecycles. Learning often overwrites itself. The Concept Library provides a stable substrate against which learning can accumulate. It does not replace learning mechanisms, but it ensures that what is learned is anchored to something permanent rather than dissolved by the next update.

The fifth thesis states that current AI systems cannot reliably share understanding. They can exchange data, outputs, or model weights, but meaning remains implicit and model-specific. The Concept Library addresses this by making meaning transferable as infrastructure. When systems reference the same immutable concepts, they can exchange interpretations and decisions in a shared semantic language, even if their internal reasoning mechanisms differ. This enables the sharing of understanding rather than merely the sharing of results.

The sixth thesis concerns locality. Intelligence that truly lives inside devices requires more than running inference at the edge. It requires continuity, semantic stability, and learning that compounds without dependence on centralized retraining. The Concept Library enables this by providing a compact, verifiable semantic layer that can be embedded locally. Systems can evolve independently while remaining

anchored to shared definitions, allowing intelligence to persist and grow without constant cloud mediation.

The seventh thesis highlights a structural inefficiency: many systems must repeatedly reconstruct meaning from scratch. Context is rebuilt, interpretations are rederived, and semantic scaffolding emerges transiently rather than being retained. The Concept Library reverses this pattern. It acts as long-term memory for identity. Semantic reconstruction becomes semantic reference, allowing computation to move forward rather than loop back.

The eighth thesis addresses accountability. As AI enters domains governed by law, safety, and liability, explanation becomes mandatory. In current architectures, explanation is often retrospective—a narrative constructed after the fact to approximate opaque internal dynamics. The Concept Library changes the nature of explanation. When meaning is external and stable, decisions can be recorded as sequences of activated concepts and resolved interpretations. Accountability becomes inspectable rather than interpretive, and explanation becomes an architectural property rather than an auxiliary tool.

The ninth thesis recognizes that regulation has transformed explainability from an optional feature into an architectural requirement. The Concept Library responds to this shift not by adding compliance layers, but by restructuring where meaning resides. When semantic identity is explicit, immutable, and persistent, explainability emerges naturally from the system's structure. Compliance is no longer a procedural overlay, but a consequence of design.

The tenth thesis states that artificial intelligence which does not understand local context cannot be trusted. Trust requires more than accuracy; it requires verifiable understanding of the situation in which a decision was made. In a Concept Library architecture, context is not inferred abstractly or reconstructed probabilistically after the fact. It is grounded in explicit concepts that define what the system understood at the moment of action. Local meaning becomes inspectable, making trust a function of shared understanding rather than blind confidence.

Taken together, these responses define the role of the Concept Library. It is not a rival to intelligence, nor a replacement for models, agents, or learning systems. It is the semantic foundation beneath them. It does not prescribe how systems must reason or act. It specifies what must remain stable if reasoning is to be accountable, learning is to be cumulative, and understanding is to be shared across systems, institutions, and time.

The shift it proposes is simple to state and difficult to ignore: intelligence can remain adaptive only if meaning is allowed to remain fixed.

Conclusion: The Choice of Foundation

The most powerful AI systems ever built still cannot reliably explain their own decisions. This is no longer a technical limitation. It is a structural crisis.

The gap between capability and accountability has become unsustainable. Regulatory frameworks demand traceability. Insurance requires stable causality. Safety-critical systems require decisions that can be inspected—not merely simulated after the fact.

The Concept Library addresses this at the architectural level. By separating immutable meaning from adaptive reasoning, it provides a semantic foundation that probabilistic inference alone cannot deliver. This is not a marginal improvement or an optimization at the edges. It is a fundamental reorientation of how intelligent systems must be built when they are expected to operate under institutional, legal, and ethical constraints.

The architecture itself is sound. Its realization, however, requires coordination.

No single organization can define the semantic foundation for transportation, healthcare, finance, and industrial systems. The concepts that enable accountability must be developed collaboratively by those who will ultimately depend on them: engineers, domain experts, regulators, insurers, and institutions responsible for long-term oversight.
The question is no longer whether such a foundation is necessary. That question has already been answered by regulation, by liability, and by operational reality.

The remaining question is how its construction begins—and who chooses to participate.

The cost of delay is not measured in missed innovation. It is measured in accidents that cannot be explained, liability that cannot be insured, and systems that fail to earn the trust required for deployment.
The foundation will be built.

The only question is whether it will be built deliberately—or improvised under crisis.

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Abstract

Current artificial intelligence architectures are optimized for guessing, not for understanding. They produce impressive results, but their decision-making is grounded in probabilistic inference, transient internal model states, and meanings that cannot be reliably inspected, verified, or preserved unchanged over time.

As AI systems move deeper into regulated, safety-critical, and legally accountable domains, this architectural foundation is no longer sufficient. Regulation, such as the EU AI Act, does not merely require good outcomes. It requires durable transparency, explainability, documentation, and the ability for human oversight to be exercised even long after a decision has been made. These requirements do not target individual algorithms or models; they target the structure of the system itself.

They presuppose that decisions are grounded in meanings that remain identifiable, inspectable, and stable regardless of how learning systems evolve over time.

The Concept Library addresses this challenge at the architectural level by separating meaning from learning and inference. It consists of individual, immutable concepts, each represented as its own standardized file. These concepts define what a system means by entities such as actors, conditions, risks, or states. They are not internal model weights or statistical representations, but explicit and durable semantic reference points to which decision-making is anchored.

This structure allows AI systems to continue learning, adapting, and improving their behavior. Learning occurs in reasoning and decision processes, not at the level of meaning itself. Meanings remain stable even as models, algorithms, and learning methods change.

As a result, AI systems do not merely produce outputs; they make decisions based on identifiable concepts in a way that can later be inspected, audited, and documented.

The Concept Library is not an add-on to existing AI models, nor an explanatory layer applied after the fact. It represents an architectural transition that becomes necessary precisely where current AI approaches reach their limits. When intelligence must be not only effective but also accountable, insurable, and compliant with regulation, meaning must be a permanent part of the system — not a retrospective interpretation.

Zusammenfassung

Die gegenwärtige KI-Architektur ist auf Vorhersage optimiert, nicht auf Verständnis. Sie erzielt beeindruckende Ergebnisse, doch ihre Entscheidungsfindung beruht auf probabilistischen Modellen, veränderlichen internen Zuständen und Bedeutungen, die weder dauerhaft überprüfbar noch über längere Zeiträume hinweg stabil dokumentierbar sind. Mit dem Einsatz von KI in regulierten, sicherheitskritischen und haftungsrelevanten Bereichen stößt diese Architektur zunehmend an ihre strukturellen Grenzen.

Regulatorische Rahmenwerke wie der EU AI Act verlangen nicht nur leistungsfähige Systeme, sondern vor allem dauerhafte Transparenz, Nachvollziehbarkeit, Dokumentierbarkeit und menschliche Aufsicht über den gesamten Lebenszyklus hinweg. Diese Anforderungen betreffen nicht einzelne Modelle oder Algorithmen, sondern die grundlegende Struktur intelligenter Systeme. Sie setzen voraus, dass Entscheidungen auf klar identifizierbaren, überprüfbaren und zeitlich stabilen Bedeutungen beruhen – unabhängig davon, wie sich Lernverfahren oder Implementierungen weiterentwickeln.

Die Concept Library adressiert diese Anforderungen auf architektonischer Ebene. Sie trennt Bedeutung konsequent vom Lernen und vom adaptiven Schlussfolgern. Die Concept Library besteht aus einzelnen, unveränderlichen Konzepten, die jeweils als eigene standardisierte Dateien definiert sind. Diese Konzepte legen fest, was das System unter Akteuren, Zuständen, Risiken oder Bedingungen versteht. Sie sind keine statistischen Repräsentationen oder Modellgewichte, sondern explizite, stabile Referenzpunkte, auf die sich jede Entscheidung bezieht.

Diese Struktur ermöglicht es KI-Systemen, weiterhin selbstständig zu lernen, sich an neue Situationen anzupassen und ihr Verhalten zu optimieren. Das Lernen findet in der Entscheidungslogik statt – nicht auf der Ebene der Bedeutung. Die Bedeutung selbst bleibt konstant, auch wenn Modelle, Algorithmen oder Trainingsmethoden wechseln. Dadurch entstehen Entscheidungen, deren semantische Grundlage dauerhaft nachvollziehbar, prüfbar und dokumentierbar ist.

Die Concept Library ist kein Zusatzmodul und keine nachträgliche Erklärungsschicht für bestehende KI-Modelle. Sie stellt einen architektonischen Übergang dar, der notwendig wird, sobald KI nicht nur leistungsfähig, sondern auch regelkonform, haftungsfähig und langfristig vertrauenswürdig sein muss. Wenn Intelligenz in institutionellen Kontexten eingesetzt wird, darf Bedeutung kein implizites Nebenprodukt mehr sein – sie muss ein expliziter, stabiler Bestandteil der Systemarchitektur werden.

Résumé

L'architecture actuelle de l'IA est optimisée pour la prédiction, et non pour la compréhension. Elle produit des résultats impressionnants, mais sa prise de décision repose sur des probabilités, des états internes changeants et des significations qui ne peuvent être inspectées, vérifiées ou préservées dans le temps. À mesure que les systèmes d'IA pénètrent dans des environnements réglementés, critiques pour la sécurité et juridiquement responsables, cette approche architecturale ne suffit plus.

Les réglementations, comme l'AI Act de l'UE, n'exigent pas seulement de bons résultats, mais aussi une transparence permanente, une explicabilité, une documentabilité et une supervision humaine, même après le déploiement. Ces exigences ne concernent pas les algorithmes ou modèles individuels, mais la structure fondamentale de l'IA. Elles nécessitent que la prise de décision soit basée sur des significations identifiables, inspectables et durables, indépendamment de l'évolution du système d'apprentissage au fil du temps.

La Concept Library (Bibliothèque de Concepts) relève ce défi à un niveau architectural. Elle sépare la signification de l'apprentissage et du raisonnement. La Concept Library se compose de concepts individuels et immuables, chacun défini dans son propre fichier standardisé. Ces concepts spécifient ce que le système entend par acteurs, conditions, risques ou états. Ils ne sont pas des poids internes de modèle ou des représentations statistiques, mais des points de référence explicites et stables auxquels les décisions de l'IA sont ancrées.

Cette structure permet à l'IA de continuer à apprendre de manière indépendante, de s'adapter à de nouvelles situations et de faire évoluer son comportement. L'apprentissage se produit dans le raisonnement et la prise de décision, et non au niveau de la signification. Les significations restent stables, même lorsque les modèles, les algorithmes et les méthodes d'apprentissage changent. Ainsi, l'IA ne produit pas seulement des réponses, mais prend des décisions basées sur des concepts identifiables, qui peuvent ensuite être examinés, audités et documentés.

La Concept Library n'est pas un module complémentaire aux modèles d'IA existants ni une couche explicative ajoutée par-dessus. C'est un changement architectural qui devient nécessaire précisément lorsque l'IA actuelle atteint ses limites. Lorsque l'intelligence doit être non seulement efficace, mais aussi responsable, assurable et conforme aux réglementations, la signification doit faire partie intégrante du système — et non être une interprétation a posteriori.

Appendix A — Proof of Concept: Semantic Decision Trace

This appendix demonstrates, in concrete terms, what it means to separate immutable meaning from adaptive reasoning.

The purpose of this example is not to present a full system, but to show that the architectural claims made in this paper are technically realizable using lightweight, inspectable structures.

No machine learning model, neural network, or probabilistic inference is shown here.

What is demonstrated is semantic traceability.

Scenario

An autonomous vehicle approaches a pedestrian crossing at dusk.

The road surface is wet. Visibility is reduced.

A decision must be made.

Semantic Activation Record (Concept Layer)

At the moment of decision, the system activates a set of predefined, immutable concepts from the Concept Library.

Concept Identifier

Concept Name

Activation State

CL-0012

pedestrian_crossing

active

CL-0047

vulnerable_road_user

active

CL-0103

reduced_visibility

active

CL-0089

low_traction_surface

active

CL-0211

collision_risk

elevated

These concepts are not inferred post hoc.

They are explicit references to stable definitions shared across systems and time.

Contextual Parameters (Interpretation Layer)

visibility_level: low

surface_condition: wet

relative_speed: moderate

braking_distance_margin: reduced

These parameters may be learned, adaptive, or probabilistic.

Their interpretation is anchored to fixed concepts.

Conflict Resolution

The system detects a semantic conflict between two operational goals:

traffic_flow_efficiency

human_safety

A safety-priority rule is triggered because the following concepts are simultaneously active:

vulnerable_road_user

reduced_visibility

elevated collision_risk

Decision Outcome

Action: controlled braking

Priority applied: safety_over_efficiency

Execution: immediate

Energy constraints are not applied, because safety-critical decisions override optimization logic by design.

Semantic Decision Log (Persistent Record)

The system records:

which concepts were active

how they were interpreted

which priority rule resolved the conflict

This record remains valid even if:

the learning model is replaced

the inference algorithm changes

the system is updated years later

Why This Constitutes Proof of Concept

This example demonstrates that:

Meaning can be explicit, inspectable, and immutable

Learning and adaptation can remain fully dynamic

Decisions can be audited by inspecting what the system understood, not how a model behaved internally

No part of this mechanism depends on:

large language models

opaque internal representations

post-hoc explanation tools

Architectural Implication

This semantic trace functions as a semantic black box:

analogous to a flight recorder

focused on meaning rather than mechanics

It enables accountability without sacrificing adaptability.

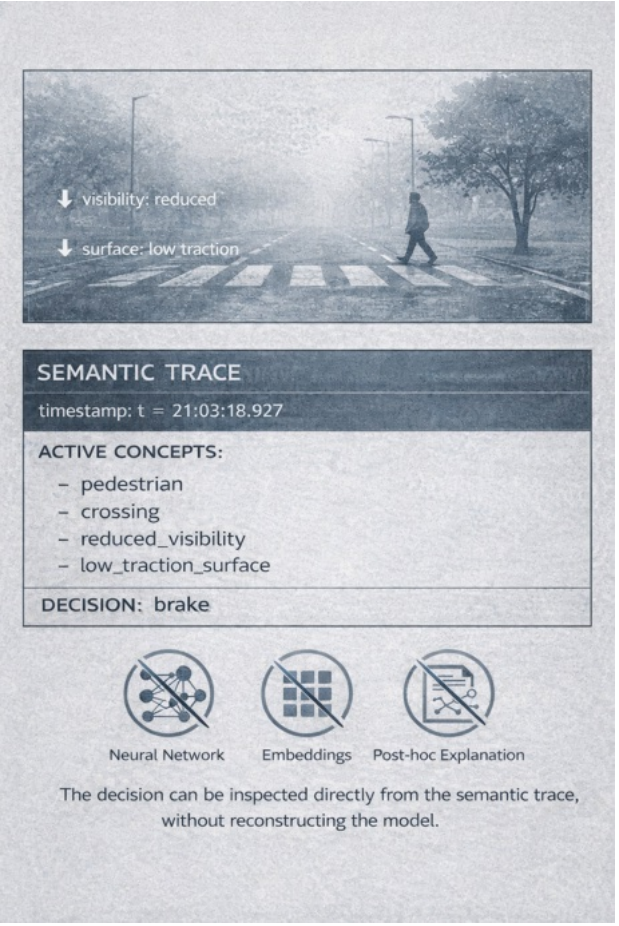
Scope and Non-Disclosure

This appendix does not disclose:

proprietary models

optimization strategies

competitive inference techniques

It demonstrates only what is necessary:

that semantic accountability is architecturally feasible.